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Pedestrian Reconstruction in Street Gaussians via
Part-Based Rigid Decomposition

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Abstract

Street Gaussians extends 3D Gaussian Splatting to dynamic urban driving scenes by decomposing each scene into a static background and per-vehicle rigid object models, achieving real-time rendering with state-of-the-art quality. Its rigid-motion assumption, however, excludes pedestrians, and later work has restored them only by reintroducing per-Gaussian skinning or deformation networks.

This thesis asks whether articulated pedestrians can be reconstructed while staying within Street Gaussians’ rigid-object design. The proposed method decomposes each pedestrian into ten approximately rigid body segments along the SMPL skeleton, represents each by an independent set of 3D Gaussians in a local frame, and transforms them into world space at render time by forward kinematics on per-instance, learnable pose residuals. The rasterizer, loss functions, and adaptive density control are unchanged: pedestrians simply contribute multiple rigid objects rather than one, with no skinning weights, no per-Gaussian blending, and no neural network at render time.

Evaluated on four Waymo scenes, the decomposition improves human-region PSNR by up to 3.69 dB where pedestrians are close, removing the limb smearing of a single rigid transform, at no measurable increase in training time. The result is a distance-dependent trade-off rather than a uniform gain: the method wins on two of the four scenes and loses on the other two, for a mean of roughly +0.8 dB. The benefit is distance-dependent: where monocular SMPL estimates become unreliable at far range, committing to a noisy skeleton scatters the segments and the decomposition falls up to 0.78 dB below the baseline. A full skinning-and-deformation method (OmniRe) remains the quality upper bound, but trains longer. The thesis additionally identifies a silent distortion-handling bug in both Street Gaussians implementations.

Part-based rigid decomposition thus recovers articulated pedestrians at the close ranges most relevant to budget-constrained deployments, while motivating a confidence-gated fallback to a single rigid object at long range as the key direction for future work.

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1 Introduction

Reconstructing dynamic three-dimensional environments from sensor data is a central problem in computer vision with direct practical impact. Autonomous driving systems require large volumes of realistic data to validate perception and planning, yet real-world collection is costly, slow, and limited to observed scenarios. High-fidelity urban digital twins address this bottleneck: after a short real capture, novel viewpoints, weather conditions, and traffic configurations can be rendered on demand. Similar representations are also relevant for city-scale digital twins, virtual production, and embodied AI training.

Two properties are essential: visual fidelity and rendering speed. Novel views must remain realistic enough for downstream models to generalize, while simulation and interactive use require tens to hundreds of frames per second. Neural Radiance Fields (NeRF) [Mil+20] largely solve fidelity but remain too slow for this regime. Because NeRF renders each pixel via many MLP evaluations along a ray, driving-scale images require massive numbers of forward passes and typically render far below real time. Urban and dynamic NeRF variants [Tan+22; Rem+22; Ost+21; Wu+23; Yan+24b] inherit or amplify this computational cost.

A separate line of work replaces implicit neural fields with explicit primitives. 3D Gaussian Splatting (3DGS) [Ker+23] models a scene as anisotropic Gaussians and renders them with a differentiable rasterizer, avoiding NeRF’s per-pixel network evaluations. This shift enables real-time rendering for complex static scenes at competitive quality. However, standard 3DGS assumes all Gaussians are fixed in world space, an assumption violated in driving scenes with moving actors; naive application therefore produces severe ghosting.

This thesis studies Street Gaussians [Yan+24a], which extends 3DGS to dynamic urban scenes while preserving high rendering efficiency. Its key idea is explicit scene decomposition: a static background point cloud plus one point cloud per tracked object. Object Gaussians are defined in local coordinates and transformed into world space at render time using tracked bounding-box poses, with pose refinement learned during training to reduce detector and tracker noise. Temporal appearance variation is modeled with 4D spherical harmonics, where SH coefficients are parameterized as Fourier series over time. On Waymo, the method reconstructs sequences in about 30 minutes on a single consumer GPU and renders at 135 FPS at 1066×1600, around two orders of magnitude faster than contemporary NeRF-based approaches while improving reconstruction quality.

The thesis has three objectives. First, it provides a technical analysis of Street Gaussians with emphasis on why it is fast: no neural network at render time, composition by point-cloud concatenation instead of compositional ray marching, and direct gradient flow from image loss to geometry and pose. The required preliminaries are developed to keep this analysis self-contained.

Second, it empirically investigates the method’s main limitation: the rigid-object assumption, which excludes pedestrians and other articulated actors. While later work such as OmniRe [Che+25] addresses this with SMPL-anchored Gaussians and linear blend skinning for close-range pedestrians — falling back to a self-supervised deformation field for far-range or template-less actors where the SMPL fit is unreliable — those gains come with per-Gaussian geometric updates, residual deformation networks, and larger parameter counts. This thesis explores a third design point: applying Street Gaussians’ rigid-object pipeline *hierarchically*, decomposing each pedestrian into ten approximately rigid body segments along the SMPL skeleton and posing them by forward kinematics on per-instance, learnable pose residuals. Each Gaussian remains rigid in its segment’s local frame and is transformed by a single rigid pose; no skinning weights, no per-Gaussian blending, and no render-time MLP are required. The experiments quantify the trade-off between this lightweight design and full skinning-based alternatives.

Third, it positions Street Gaussians within the broader landscape roughly two years after publication, focusing on two trends: feed-forward, generalizable 3D Gaussian prediction that avoids per-scene optimization, and annotation-free decomposition where self-supervised signals replace the 3D bounding box supervision Street Gaussians depends on.

The remainder of this thesis is organized as follows. Chapter 2 covers the mathematical and conceptual background: NeRF, 3DGS, spherical harmonics, and the failure mode of static 3DGS in dynamic scenes. Chapter 3 details Street Gaussians, including scene decomposition, 4D SH, rendering, and training. Chapter 4 reviews results from the original paper. Chapter 5 presents the pedestrian extension — the proposed part-based rigid decomposition and comparisons against rigid and full-skinning baselines; the Appendix documents a calibration fix found in both Street Gaussians implementations. Chapter 6 concludes.

2 Background and Preliminaries

This chapter develops the technical foundations required to understand Street Gaussians. Section 2.1 introduces Neural Radiance Fields and analyzes the source of their computational cost. Section 2.2 presents 3D Gaussian Splatting as an explicit alternative and explains why the change in representation yields orders-of-magnitude speedups. Section 2.3 covers spherical harmonics, the mechanism both methods use to encode view-dependent color. Section 2.4 examines why 3D Gaussian Splatting, despite its efficiency, cannot directly handle dynamic scenes — motivating the decomposition strategy of Street Gaussians presented in Chapter 3.

2.1 Neural Radiance Fields

Neural Radiance Fields (NeRF) [Mil+20] represent a 3D scene as a continuous function

$$f_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma), \quad (2.1)$$

where $\mathbf{x} \in \mathbb{R}^3$ is a spatial position, $\mathbf{d} \in \mathbb{S}^2$ is a viewing direction, $\mathbf{c} \in \mathbb{R}^3$ is the emitted radiance (color), and $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at that point. The function f_θ is parameterized by a multi-layer perceptron (MLP) with weights θ , typically eight fully connected layers of 256 hidden units each.

Volumetric rendering. To render a pixel, NeRF emits a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ from the camera origin $\mathbf{o}$ through the pixel and integrates radiance along it. The expected color of the ray is given by the volumetric rendering integral:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{d}) dt, \quad T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{r}(s)) ds\right), \quad (2.2)$$

where $T(t)$ is the accumulated transmittance from the near bound t_n to t , modeling the probability that the ray has not been absorbed by earlier media. The integral is approximated in practice by discretization: N sample points are drawn along the ray, the MLP is evaluated at each, and the integral is replaced by a numerical quadrature rule that reduces to compositional alpha blending:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N T_i (1 - \exp(-\sigma_i \delta_i)) \mathbf{c}_i, \quad T_i = \exp\left(-\sum_{j=1}^{i-1} \sigma_j \delta_j\right), \quad (2.3)$$

where $\delta_i = t_{i+1} - t_i$ is the distance between consecutive samples. The entire pipeline is differentiable, permitting θ to be optimized from multi-view images by minimizing a photometric loss between rendered and observed pixels.

The computational bottleneck. The cost of this scheme is dominated by MLP evaluations: rendering a full $H \times W$ image requires $H \times W \times N$ forward passes through f_θ , roughly 2.2×10^8 per frame at the resolution used in Street Gaussians (1066×1600 , $N = 128$). A substantial body of work attacks this cost by replacing or augmenting the MLP with explicit data structures: a hash grid in Instant-NGP [Mül+22], a network-free voxel grid of SH coefficients in Plenoxels [Fri+22], or the anti-aliased Mip-NeRF 360 / Zip-NeRF line that defines the benchmark for large outdoor scenes [Bar+22; Bar+23]. Even so, these variants do not reach real-time rendering at such resolutions: the baselines compared in the Street Gaussians paper achieve 0.21 to 0.68 FPS, with training taking hours to days.

2.2 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) [Ker+23] replaces the implicit MLP representation with an explicit set of geometric primitives. Instead of a function that can be queried anywhere in space, the scene is represented directly as a finite collection of 3D Gaussians, each carrying its own appearance parameters. Rendering proceeds by projecting these primitives onto the image plane and rasterizing them — a pipeline that maps naturally onto GPU tile-based rasterization hardware and avoids per-pixel network evaluation entirely.

The Gaussian primitive. A single 3D Gaussian is defined by its position $\boldsymbol{\mu} \in \mathbb{R}^3$ (the mean) and a covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{3 \times 3}$ that describes its shape and orientation. The spatial falloff of the Gaussian is

$$G(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (2.4)$$

For $\boldsymbol{\Sigma}$ to represent a valid covariance, it must be symmetric and positive semi-definite. Optimizing a general 3×3 matrix under these constraints is numerically fragile, so 3DGS instead parameterizes $\boldsymbol{\Sigma}$ via a scaling matrix $\mathbf{S} = \text{diag}(s_1, s_2, s_3)$ and a rotation matrix $\mathbf{R}$ derived from a unit quaternion $\mathbf{q}$:

$$\boldsymbol{\Sigma} = \mathbf{R} \mathbf{S} \mathbf{S}^\top \mathbf{R}^\top. \quad (2.5)$$

This factorization is automatically positive semi-definite for any values of $\mathbf{q}$ and s_i , removing the need for constrained optimization. In practice the scales are stored in log space so that $s_i > 0$ is enforced by exponentiation.

Each Gaussian additionally carries two appearance parameters: an opacity $\alpha \in [0, 1]$ that modulates its contribution during compositing, and a set of spherical harmonics coefficients $\{z_{m,\ell}\}$ that encode view-dependent color (discussed in Section 2.3). The

full parameter set per Gaussian is thus $\{\boldsymbol{\mu}, \mathbf{q}, \mathbf{s}, \alpha, \mathbf{z}\}$ — typically around 60 scalar values depending on the SH degree.

Projection to the image plane. To render, each 3D Gaussian must be projected onto the 2D image plane. Given a camera with extrinsic matrix $\mathbf{W}$ (world-to-camera transform) and intrinsic matrix $\mathbf{K}$, the mean projects straightforwardly:

$$\boldsymbol{\mu}' = \mathbf{K}\mathbf{W}\boldsymbol{\mu}. \quad (2.6)$$

The covariance, however, does not project linearly under perspective projection. 3DGS uses the elliptical-weighted-average (EWA) splatting framework of Zwicker et al. [Zwi+02], which formalizes the perspective projection of anisotropic Gaussian kernels together with a screen-space low-pass filter that band-limits them: the perspective projection is linearized around each Gaussian’s mean via its Jacobian $\mathbf{J}$, and the 2D covariance is computed as

$$\boldsymbol{\Sigma}' = \mathbf{J}\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top \mathbf{J}^\top. \quad (2.7)$$

Here $\mathbf{W}$ denotes the linear (rotational) part of the world-to-camera transform — distinct from the full extrinsic applied to the mean above, since a covariance transforms only under the linear part — and the upper-left 2×2 block of the result is retained. The resulting 2×2 matrix $\boldsymbol{\Sigma}'$ defines an elliptical footprint in image space. Each Gaussian becomes, from the rasterizer’s perspective, an oriented ellipse with a Gaussian falloff.

Tile-based rasterization and alpha blending. Rendering then follows the standard graphics pipeline. The image is divided into tiles (typically 16×16 pixels). Each projected Gaussian is assigned to the tiles its footprint overlaps, and within each tile the Gaussians are sorted by depth. For each pixel, the final color is computed by front-to-back alpha compositing:

$$C = \sum_{i=1}^N \mathbf{c}_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (2.8)$$

where α_i is the primitive opacity multiplied by the 2D Gaussian evaluated at the pixel location, and $\mathbf{c}_i$ is the view-dependent color from the spherical harmonics. This compositing rule is formally identical to NeRF’s quadrature (Section 2.1), but the summation is over primitives intersected by the pixel rather than samples along a ray — and critically, no neural network is evaluated in the inner loop.

Why this is fast. With no per-sample MLP evaluation, the rasterization stage mapping directly onto GPU tile-based hardware, and gradients flowing through the dif-

ferentiable rasterizer without a deep network, 3DGS can train scenes in minutes and render them at hundreds of frames per second — orders of magnitude faster than NeRF at equivalent resolution.

Adaptive density control. The efficiency of 3DGS relies on allocating Gaussians where the scene actually needs them. Starting from a sparse SfM point cloud, 3DGS runs a periodic densification-and-pruning procedure driven by the view-space positional gradient $\|\nabla_{\mu}\mathcal{L}\|$: a large gradient signals that a Gaussian is failing to represent its local region. Two failure modes are distinguished by size. A *small* Gaussian with a large gradient is under-covering nearby geometry; it is *cloned*, and the two copies diverge under optimization to fill the gap. A *large* Gaussian with a large gradient is too coarse for local detail; it is *split* into two smaller children sampled from the original distribution. Gaussians whose opacity falls below a threshold are pruned periodically. The result is a scene-adaptive allocation: complex regions accumulate many small Gaussians while large smooth surfaces are covered by few large ones. Street Gaussians inherits this mechanism for the background model, adding one step — object Gaussians that drift outside their assigned bounding box are pruned after each densification — which keeps object models compact but, as Chapter 5 shows, also becomes a failure mode for non-rigid objects whose limbs leave the box.

2.3 Spherical Harmonics for View-Dependent Color

Both NeRF and 3DGS must encode the fact that real surfaces look different from different angles — specular highlights move with the viewer, subsurface scattering varies with incidence angle, and so on. NeRF accomplishes this by passing the viewing direction $\mathbf{d}$ as an additional input to its MLP. 3DGS needs a different mechanism: spherical harmonics.

Spherical harmonics are real-valued basis functions $Y_{\ell}^m : \mathbb{S}^2 \rightarrow \mathbb{R}$ on the unit sphere, indexed by a degree ℓ and order m with $-\ell \leq m \leq \ell$ — the spherical analogue of the Fourier basis. Any sufficiently smooth function on the sphere expands as

$$f(\mathbf{d}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} z_{m,\ell} Y_{\ell}^m(\mathbf{d}), \quad (2.9)$$

and truncating at degree $\ell_{\max}$ yields $(\ell_{\max} + 1)^2$ basis functions. Each Gaussian stores one coefficient set per RGB channel, and view-dependent color is a near-free dot product between the stored coefficients and the basis values at $\mathbf{d}$.

Degree truncation. The choice of $\ell_{\max}$ trades expressiveness against storage: degree 1 captures smooth directional variation, higher degrees capture sharp effects like

metallic highlights. The original 3DGS paper uses degree 3; Street Gaussians reduces this to degree 1, arguing that urban scenes have weak view-dependent effects and that lower degrees resist overfitting to training-view lighting. This cuts per-Gaussian appearance storage fourfold and is a non-trivial source of the method’s efficiency advantage.

2.4 Why Static 3DGS Fails for Dynamic Scenes

The efficiency of 3DGS rests on a strong implicit assumption: the scene is static. Every Gaussian has a fixed position μ in world coordinates, and the rasterizer treats the scene as time-invariant. This assumption is directly violated by driving data, in which vehicles move along the road, pedestrians cross intersections, and the ego camera itself changes its relationship to these moving objects frame by frame.

The ghosting failure mode. Consider a car that is observed in frame 1 at world position $\mathbf{p}_1$ and in frame 10 at position $\mathbf{p}_2$. Because 3DGS has no mechanism to associate Gaussians with time, the only way it can reconcile these observations within a single world-space point cloud is to place *some* Gaussians at $\mathbf{p}_1$ and *some* at $\mathbf{p}_2$ (to explain either frames), each with intermediate opacity. At every frame, both sets project to the image — one in the correct location, one in the wrong one. The result is the “ghosting” artifact visible in the 3DGS column of the Street Gaussians paper’s qualitative comparisons: transparent copies of moving objects smeared along their trajectories, and severe reconstruction failure in the pixels those trajectories cover. The training loss has no means to resolve this; any arrangement that places mass along the trajectory is roughly equivalent under a per-frame photometric objective.

The decomposition insight. Naive fixes, like per-frame Gaussian positions, or conditioning positions on time via an MLP either discard cross-frame structure or reintroduce the per-sample network evaluation that 3DGS was designed to avoid. Street Gaussians instead exploits a specific property: most of the world is static, and the dynamic portion is a small number of rigid objects whose motion is already tracked by the perception stack. Static content stays in a world-space point cloud while each tracked object gets its own cloud in a local frame, moved into world space at render time by its rigid pose, preserving 3DGS efficiency while accommodating the dominant form of motion. Chapter 3 develops this pipeline in detail.

The cost is that the design assumes rigid dynamics: a moving car is well-approximated by a rigid transform of its point cloud, but a walking pedestrian is not, because limbs articulate relative to the torso. The consequences of violating this assumption are the subject of Chapter 5.

3 Street Gaussians: Method

This chapter presents the Street Gaussians method in detail, covering the scene decomposition, the background and foreground models, the rendering pipeline, training, and the sources of the method’s efficiency advantage.

3.1 Scene Decomposition

The core observation motivating Street Gaussians is that dynamic urban scenes are not uniformly dynamic. The vast majority of the scene, like roads, buildings, trees, parked cars and sky is static over the duration of a typical driving sequence. The dynamic portion consists almost entirely of a small number of rigid objects (moving vehicles) whose 3D positions over time are already being estimated by the perception stack of any self-driving platform. The tracker output, although noisy, provides strong prior information about where each dynamic object is at each frame.

Street Gaussians exploits this structure by decomposing the scene into components according to their motion:

1. A background model: a single set of 3D Gaussians in world coordinates, representing everything that does not move across frames.
2. Foreground object models: one set of 3D Gaussians per tracked moving vehicle, each defined in the object’s own local coordinate frame.

At any given frame, the world-space state of the scene is obtained by keeping the background Gaussians where they are and transforming each object’s Gaussians from its local frame into world space using that object’s tracked pose at the current timestep. The combined set of world-space Gaussians is then rendered by the standard 3DGS rasterizer. This design has three immediate consequences:

Motion is absorbed entirely into the pose transformations. The Gaussian parameters — position, rotation, scale, opacity, color — are time-invariant within each object’s local frame; only the per-frame rigid transforms ($\mathbf{R}_t, \mathbf{T}_t$) carry temporal information, so a single set of Gaussians per object handles the entire sequence.

Composition is free. All world-space Gaussians — static and transformed-dynamic — are concatenated into a single list and handed to the unmodified 3DGS rasterizer, requiring no new rendering algorithm and preserving the efficiency advantage of 3DGS end-to-end.

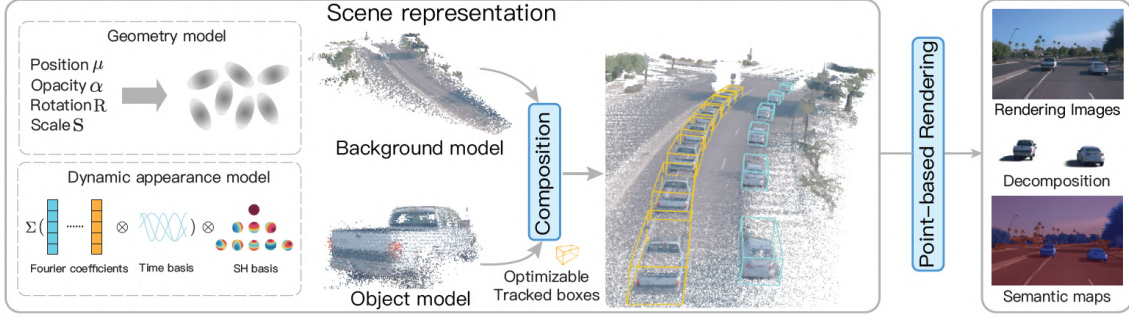


Figure 1 Overview of the Street Gaussians representation. A dynamic urban scene is modeled as a set of point-based background and foreground objects with optimizable tracked vehicle poses. Each point carries a 3D Gaussian [Ker+23] (position, opacity, and a covariance from rotation and scale) for geometry; background points use a spherical harmonics appearance model, while foreground points use a dynamic spherical harmonics model. The explicit point-based representation makes composition of the separate models trivial, enabling real-time rendering of high-quality images and (optional) semantic maps, as well as decomposition of foreground objects for editing. Figure from Yan et al. [Yan+24a].

Editing is enabled as a side effect. Because each object is an independent point cloud with its own pose, objects can be removed, translated, or swapped at inference time without retraining, making the scene usable as a closed-loop driving simulator in the spirit of UniSim [Yan+23].

This static-background-plus-per-object-node decomposition is inherited from Neural Scene Graphs [Ost+21], with each node instantiated as a set of Gaussians rather than a radiance-field MLP. Extending 3DGS to motion has since split into two families. The *deformation-field* family keeps one canonical Gaussian set and predicts per-frame motion via an MLP or spatio-temporal field [Lui+24; Yan+24c; Wu+24; Li+24a]. Street Gaussians belongs instead to the *scene-graph / object-decomposition* family, which keeps motion explicit; its peers add a LiDAR-prior composite graph [Zho+24c], physically-constrained motion robust to noisy boxes [Zho+24a], or a box-free periodically-vibrating primitive [Che+23].

3.2 Background Model

The background model is essentially standard 3D Gaussian Splatting applied to the static portion of the scene. Each background point carries the full set of 3DGS parameters: position $\mu_b \in \mathbb{R}^3$, covariance $\Sigma_b = \mathbf{R}_b \mathbf{S}_b \mathbf{S}_b^\top \mathbf{R}_b^\top$, opacity $\alpha_b \in [0, 1]$, and spherical harmonics coefficients $\mathbf{z}_b$ for view-dependent color. The SH degree is set to 1 rather than the 3DGS default of 3, justified by the weaker view-dependent effects of outdoor urban scenes and the reduced risk of overfitting.

One addition extends beyond vanilla 3DGS: each background point also carries a *semantic logit* $\beta_b \in \mathbb{R}^M$, where M is the number of semantic classes. This logit is rendered through the same alpha-compositing mechanism as color and trained against 2D predictions from Video K-Net, enabling the segmentation application discussed in Chapter 4. The background model is optimized directly in world coordinates with no pose transformations.

3.3 Foreground Object Model

The foreground object model is where the dynamic handling actually happens. Each tracked moving vehicle is represented by its own independent point cloud, stored in a local coordinate frame attached to the object — not in world coordinates.

3.3.1 Local Coordinates and Pose Transformation

For an object o , each Gaussian has its position μ_o and rotation $\mathbf{R}_o$ expressed relative to a coordinate frame anchored at the object’s center. At a given frame t , the object’s tracked pose $(\mathbf{R}_t, \mathbf{T}_t)$ maps the local frame into world coordinates:

$$\mu_w = \mathbf{R}_t \mu_o + \mathbf{T}_t, \quad \mathbf{R}_w = \mathbf{R}_t \mathbf{R}_o. \quad (3.1)$$

The world-space covariance Σ_w follows from the covariance factorization with world-space rotation $\mathbf{R}_w$ and unchanged scale $\mathbf{S}_o$. A vehicle driving down the road has the same local-frame point cloud at every timestep; only the tracked pose slides the entire cloud along the vehicle’s trajectory (Figure 2).

The tracker poses that come from off-the-shelf detectors are noisy — bounding box centers are inaccurate by tens of centimetres, orientations are imprecise — and these errors manifest directly as rendering artifacts. Section 3.5 describes how this is handled by treating the poses as partially learnable.

3.3.2 4D Spherical Harmonics for Time-Varying Appearance

The rigid pose transformation handles changes in object position and orientation, but a moving vehicle’s appearance also changes over time in ways pure rigid motion cannot explain: shadows shift as the car passes under a tree, illumination changes near streetlights, specular highlights move as its orientation relative to the sun changes. Static SH coefficients model variation with viewing direction — a different degree of freedom from time.

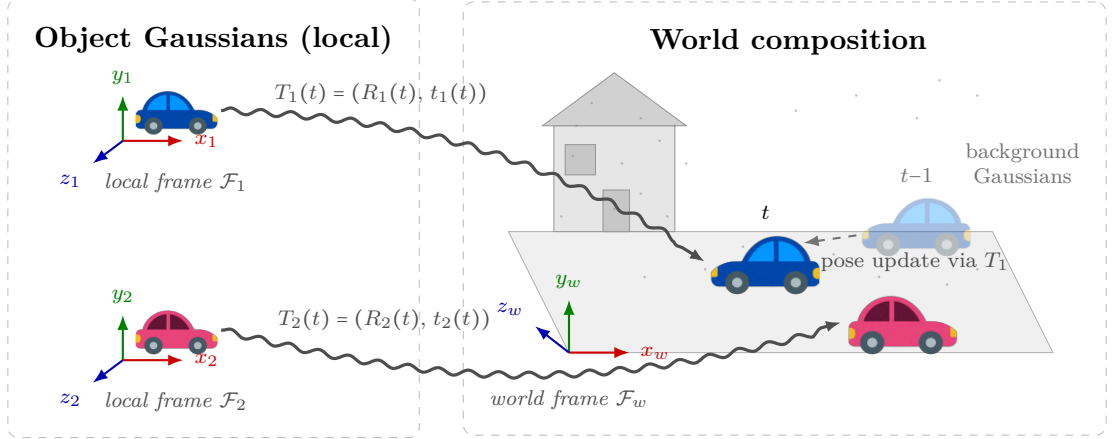


Figure 2 Pose transformation. Each dynamic object i has time-invariant Gaussians in its own frame $\mathcal{F}_i$ (left), placed into world space at frame t by the rigid transform $T_i(t) \in \text{SE}(3)$ (right). Only T_i varies with t ; the ghosted blue car shows the same Gaussians one frame earlier.

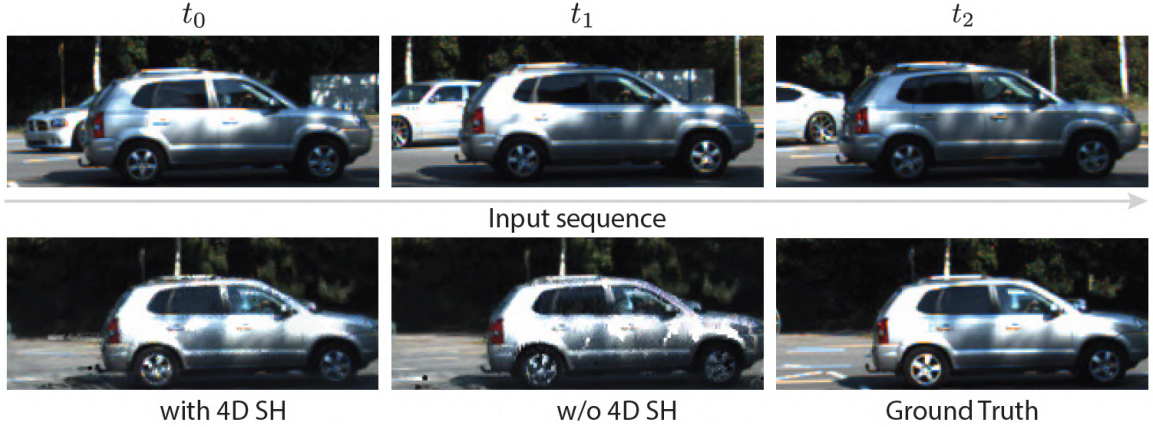


Figure 3 Effect of 4D spherical harmonics. The same vehicle is shown across several timesteps as illumination, shadows, and specular highlights change. Static SH coefficients cannot represent this variation. Figure from Yan et al. [Yan+24a].

Storing separate SH coefficients per frame would multiply per-Gaussian memory by the sequence length N_t — on the order of $200\times$ for a standard Waymo sequence. Street Gaussians instead represents each SH coefficient as a short Fourier series over time: each scalar $z_{m,\ell}$ is replaced by k Fourier coefficients $\mathbf{f} \in \mathbb{R}^k$, recovered at frame t via:

$$z_{m,\ell}(t) = \sum_{i=0}^{k-1} \mathbf{f}_i \cos\left(\frac{i\pi}{N_t}t\right). \quad (3.2)$$

With $k = 5$, storage grows by a factor of 5 rather than N_t , while still capturing the smooth temporal variation that physical lighting produces.

A limitation follows from indexing appearance by time rather than pose: the learned lighting is baked to the object’s *training* trajectory. When an object is relocated for editing — say, advanced 2 m along the road — its 4D SH still replays the shadows and

Table 1 Per-Gaussian parameter count under three appearance models. Non-appearance parameters (position $\boldsymbol{\mu}$, rotation quaternion $\mathbf{q}$, scale $\mathbf{s}$, opacity α) are identical across all three and listed once. Appearance parameters are shown separately. SH degree $\ell_{\max} = 1$, yielding 4 basis functions per channel and 12 scalars per static SH set; sequence length $N_t = 200$ frames (matching the thesis’s training scenes); Fourier truncation $k = 5$.

	Static SH <i>(background, 3DGS)</i>	Per-frame SH <i>(naive baseline)</i>	4D SH (Fourier) <i>(Street Gaussians)</i>
Geometry: $\boldsymbol{\mu}, \mathbf{q}, \mathbf{s}$	10	10	10
Opacity: α	1	1	1
Appearance (SH)	12	$12 \cdot N_t = 2,400$	$12 \cdot k = 60$
Total per Gaussian	23	2,411	71
Appearance vs. static	1×	200×	5×

highlights from where it was at each timestep, so it shows the shading of having passed a tree it now reaches earlier. The appearance is correct only along the trajectory it was trained on.

3.3.3 Object Semantics

Because all Gaussians belonging to an object represent the same class, each object model stores a single scalar semantic label β_o — the tracker-provided class — converted to a one-hot vector over M classes at render time for compatibility with the background’s M -dimensional logits during compositing. This saves memory and prevents the optimizer from inventing inconsistent semantics for Gaussians known a priori to belong to a single object.

3.4 Rendering Pipeline

Given a target frame t and camera $(\mathbf{W}, \mathbf{K})$, rendering proceeds as follows. The Fourier series (Eq. (3.2)) is first evaluated per object to recover SH coefficients at t ; each object cloud is then transformed to world space via Eq. (3.1); and all clouds (background and dynamic) are concatenated into a single primitive list. Finally, EWA splatting projects every Gaussian onto the image plane and the standard 3DGS tile-based rasterizer composites them front-to-back. The first three steps are lightweight per-frame pre-processing; only the last two touch every pixel, and both are the unmodified 3DGS rasterizer — which is what lets Street Gaussians inherit its rendering speed almost entirely.

Gaussians are poorly suited to the effectively infinite distance of the sky: large Gaussians in the upper scene produce boundary artifacts and waste optimization capacity. Street Gaussians instead uses a separate *sky cubemap* — a learnable 1024×1024-per-face texture mapping viewing direction to sky color. The rasterizer produces a Gaussian color image $\mathbf{C}_g$ and opacity image $\mathbf{O}_g$; the cubemap contributes wherever the Gaussian layer is transparent:

$$\mathbf{C} = \mathbf{C}_g + (1 - \mathbf{O}_g) \mathbf{C}_{\text{sky}}. \quad (3.3)$$

3.5 Training

Training jointly optimizes background Gaussian parameters, per-object Gaussian parameters, the sky cubemap, and per-frame pose residuals.

Tracked pose optimization. Rather than trusting tracker poses blindly or optimizing them from scratch (which is under-constrained), Street Gaussians adds small learnable residuals on top of the tracker output:

$$\mathbf{R}'_t = \mathbf{R}_t \Delta \mathbf{R}_t, \quad \mathbf{T}'_t = \mathbf{T}_t + \Delta \mathbf{T}_t, \quad (3.4)$$

where $\Delta \mathbf{T}_t \in \mathbb{R}^3$ is a free translation correction and $\Delta \mathbf{R}_t$ is built from a single learnable yaw offset $\Delta \theta_t$. Vehicles do not pitch or roll, so restricting the rotational correction to yaw reduces parameters and prevents degenerate solutions.

Because the pose transformation sits directly in the rendering path, gradients from the pixel loss flow through the rasterizer into the residuals without traversing any intermediate neural network, making pose correction both fast and precise.

Point cloud initialization. Because a point cloud can only represent regions where points actually exist, initialization matters more for explicit representations than for implicit ones. Street Gaussians uses three sources of initial points:

LiDAR aggregation provides dense, metrically accurate coverage within sensor range by aggregating per-frame scans into world coordinates. *Structure-from-Motion* (COLMAP, with moving objects masked) recovers points for distant regions beyond LiDAR range. *Object points* are LiDAR returns inside each tracked bounding box, transformed into the object’s local frame. The background combines both sources after voxel down-sampling and visibility filtering.

Loss function. The total training loss is a weighted sum of five terms:

$$\mathcal{L} = \mathcal{L}_{\text{color}} + \lambda_1 \mathcal{L}_{\text{depth}} + \lambda_2 \mathcal{L}_{\text{sky}} + \lambda_3 \mathcal{L}_{\text{sem}} + \lambda_4 \mathcal{L}_{\text{reg}}, \quad (3.5)$$

with $\lambda_1 = 0.01$, $\lambda_2 = 0.05$, $\lambda_3 = 0.1$, $\lambda_4 = 0.1$.

Color reconstruction $\mathcal{L}_{color}$. The standard 3DGS photometric loss — a convex combination of L1 pixel error and D-SSIM ($\lambda_{SSIM} = 0.2$).

Depth supervision $\mathcal{L}_{depth}$. An L1 loss between rendered depth and sparse LiDAR depth (excluding the 5% largest-error pixels as outliers), critical in weakly textured regions where photometric loss is geometrically underdetermined.

Sky supervision $\mathcal{L}_{sky}$. A binary cross-entropy loss matching rendered opacity $\mathbf{O}_g$ to a predicted sky mask $\mathbf{M}_{sky}$, preventing Gaussians from growing into the sky and keeping the cubemap boundary clean:

$$\mathcal{L}_{sky} = - \sum [(1 - \mathbf{M}_{sky}) \log \mathbf{O}_g + \mathbf{M}_{sky} \log(1 - \mathbf{O}_g)]. \quad (3.6)$$

Semantic supervision $\mathcal{L}_{sem}$. A per-pixel softmax cross-entropy loss against Video K-Net predictions, backpropagated only to the semantic logits β (following Panoptic Lifting [Siddiqui2023]) so noisy predictions cannot corrupt geometry.

Opacity regularization $\mathcal{L}_{reg}$. An entropy loss on foreground accumulated opacity that pushes each object pixel toward fully opaque or transparent, reducing “floater” artifacts visible when objects are extracted for editing:

$$\mathcal{L}_{reg} = - \sum [\mathbf{O}_{obj} \log \mathbf{O}_{obj} + (1 - \mathbf{O}_{obj}) \log(1 - \mathbf{O}_{obj})]. \quad (3.7)$$

Training runs for 30,000 iterations with Adam; pose residuals and the sky cubemap use exponentially decaying learning rates. Adaptive density control follows Chapter 2, including the bounding-box pruning of object Gaussians introduced there.

3.6 Efficiency Summary

Street Gaussians achieves 135 FPS and 30-minute training on a single GPU against 0.2–0.7 FPS and multi-hour training for NeRF competitors. Four design choices compound to produce this two-orders-of-magnitude advantage: no MLP at render time (SH dot products replace N per-pixel network queries), compositional rendering by concatenation rather than per-component volumetric integration, 4D SH that holds temporal appearance to a $5\times$ overhead instead of $N_t\times$, and direct gradient flow from pixel loss to pose residuals through the rasterizer.

4 Evaluation in the Original Paper

This chapter summarises the experimental evidence reported in the original Street Gaussians paper. The aim is not to re-report every number but to surface the findings that the part-based pedestrian decomposition in Chapter 5 attempts to address.

4.1 Setup

Evaluation uses three benchmarks. The Waymo Open Dataset [Sun+20] is the main testbed, rendered at 1066×1600 with 3D tracks from an off-the-shelf detector-tracker; KITTI [Geiger2012] and Virtual KITTI 2 [Cabon2020] are evaluated at 375×1242 across three train/test splits (75%, 50%, 25%) using ground-truth tracklets.

Four baselines span the design space. *3DGS* [Ker+23] is a static explicit Gaussian model with no object decomposition, included as a fast but dynamics-agnostic reference. *NSG* [Ost+21] and *MARS* [Wu+23] are NeRF-based scene-graph methods that decompose the scene into explicit objects but rely on ground-truth tracks. *EmerNeRF* [Yan+24b] is self-supervised and uses no tracks at all, separating static and dynamic content implicitly.

Quality is measured with the standard PSNR, SSIM, and LPIPS [Zhang2018] on the full image. Because the methods differ mostly in the small image regions covered by moving objects, the paper additionally reports $PSNR^*$ — PSNR restricted to the $1.5\times$ -expanded 2D bounding boxes of moving objects — which isolates quality exactly where decomposition matters. Rendering speed is reported as FPS at full resolution.

4.2 Quantitative Comparisons

Two findings from Table 2 matter for the rest of this thesis. First, the gain concentrates on moving objects: Street Gaussians leads EmerNeRF by 3.74 dB PSNR on Waymo but by 3.69 dB $PSNR^*$ against MARS on moving objects — the explicit per-object decomposition pays off precisely where decomposition is the point. Second, it outperforms NSG and MARS while using noisy tracker poses rather than the ground-truth tracks they consume, confirming that pose optimization more than compensates for tracker noise. The two-order-of-magnitude FPS advantage (135 vs. 0.21–0.68) follows directly from the explicit representation and is what makes the downstream applications interactive.

Table 2 Quantitative comparison from Yan et al. [Yan+24a]. Top: full results on Waymo Open at 1066×1600 . Bottom: PSNR $\uparrow$ on KITTI and Virtual KITTI 2 at three train/test splits. PSNR* denotes PSNR on moving objects. Best per column in **bold**; 3DGS FPS is for static-only rendering.

Waymo	3DGS	NSG	MARS	EmerNeRF	SG
PSNR $\uparrow$	29.64	28.31	29.75	30.87	34.61
SSIM $\uparrow$	0.918	0.862	0.886	0.905	0.938
LPIPS $\downarrow$	0.117	0.346	0.264	0.133	0.079
PSNR* $\uparrow$	21.25	24.32	26.54	21.67	30.23
FPS $\uparrow$	205	0.47	0.68	0.21	135

Method	KITTI (PSNR $\uparrow$)			Virtual KITTI 2 (PSNR $\uparrow$)		
	75%	50%	25%	75%	50%	25%
3DGS	19.19	19.23	19.06	21.12	21.11	20.84
NSG	21.53	21.26	20.00	23.41	23.23	21.29
MARS	24.23	24.00	23.23	29.79	29.63	27.01
Ours	25.79	25.52	24.53	30.10	29.91	28.52

4.3 Ablations

Table 3 isolates four design choices. *Tracked pose optimization* is by far the largest contributor, accounting for 2 dB of PSNR* while barely affecting full-image PSNR. Pose errors live entirely in the few pixels covered by moving objects. Training with ground-truth poses scores *below* the optimized result, showing that image-evidence-driven residuals can exceed annotation quality. *4D spherical harmonics* contribute ≈ 1 dB of PSNR*, mostly by removing temporally inconsistent shading on vehicles passing through shadows. *LiDAR initialization* adds 0.6 dB in weakly textured regions where photometric loss is geometrically underdetermined. *Opacity regularization* is absent from the headline table because it does not affect composed-rendering metrics, but the supplementary material shows it removes the semi-transparent halos at object silhouettes when vehicles are extracted for simulation.

The first two findings carry directly into Chapter 5: per-instance pose refinement is reused for pedestrian segments, and the SH degree is inherited rather than re-tuned.

4.4 Applications

Three downstream applications share a single structural property: each moving vehicle is a self-contained point cloud with its own tracked pose, so manipulating it

Table 3 Ablation study on Waymo [Yan+24a]. Metrics are averaged over all Waymo sequences; PSNR* denotes PSNR on moving objects and “opt.” denotes pose optimization. Each row removes or replaces one component of the complete model.

	PSNR $\uparrow$	PSNR* $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours w/o LiDAR	34.02	29.53	0.934	0.087
Ours w/o 4D SH	34.36	29.27	0.937	0.081
Ours w/o pose opt.	34.18	28.24	0.935	0.081
Ours w/ GT pose	34.61	29.84	0.937	0.080
Complete model	34.61	30.23	0.938	0.079

reduces to a pose or point-cloud operation at inference time. *Scene editing* (Figure 5: rotation, translation to a novel lane, vehicle swapping) needs no retraining and runs at full 135 FPS — the property systems such as HUGSIM exploit to lift captured images into a real-time, closed-loop driving simulator [Zho+24b]. *Object decomposition* is trivial because every Gaussian belongs to exactly one layer. *Semantic segmentation* achieves 58.81 mIoU, exceeding the Video K-Net 2D predictions (57.94) it is supervised by, because 3D aggregation forces each Gaussian to a label consistent across all views. NeRF-based alternatives offer none of these for free at inference time.

4.5 Limitations

The paper names three limitations, each relevant to the thesis.

Rigid-only dynamics. The method cannot handle non-rigid motion. Pedestrians, cyclists, and animals — all present in realistic driving scenes and all required in a full simulation environment — are out of scope. This is the direct subject of Chapter 5. *Tracker dependency.* Pose optimization can correct small tracker errors but cannot recover from missed detections; vehicles the tracker drops entirely are absorbed into the background as ghosting. Method quality is bounded by the recall of the upstream perception stack. *Per-scene optimization.* Each new scene requires 30 minutes of training from scratch, with no knowledge transfer between scenes — a fundamental constraint of the explicit representation paradigm. A fourth limitation is implicit: the method assumes a pre-existing 3D detection and tracking pipeline. For data without such tracker outputs (casual dashcam footage, historical records), Street Gaussians cannot be applied without first running a full perception pipeline.



Figure 4 Qualitative comparison on the Waymo dataset. NSG, MARS, and EmerNeRF produce ghosting and blur in regions with moving objects, whereas the explicit per-object decomposition of Street Gaussians yields sharp dynamic regions [Yan+24a].



Figure 5 Scene editing enabled by the explicit object decomposition. Each vehicle is a self-contained point cloud with its own tracked pose, so it can be rotated, translated to a novel lane, or swapped with another vehicle at inference time without retraining. [Yan+24a].

5 Implementation Contributions and Future Directions

This chapter presents the implementation contributions of the thesis and discusses broader future directions for Street Gaussians-style methods. The calibration investigation in Appendix Section A.1 spans both Street Gaussians implementations — the official release [Yan+24a] and the Nerfstudio-based community port *street-gaussians-ns* by LightwheelAI — where the same distortion-handling bug was found and fixed. The final pedestrian experiments are conducted on the DriveStudio repository, extended with that calibration fix and the part-based pedestrian representation developed in this thesis. Section 5.1 presents the main extension of the thesis: a part-based rigid decomposition of pedestrians that applies the Street Gaussians representation hierarchically to body segments rather than treating each pedestrian as a single rigid object. Section 5.2 situates the method within the broader research landscape and discusses concrete directions for future work.

5.1 Part-Based Rigid Decomposition for Pedestrians

Street Gaussians excludes pedestrians from its scope, citing rigid motion as a fundamental limitation [Yan+24a]. OmniRe [Che+25], the current state of the art, addresses this gap (28.15 dB PSNR on Waymo human regions vs. 16.83 dB for rigid Street Gaussians) by extending the scene-graph design with two new node types and *routing each actor by distance and template fit*: close-range walking pedestrians, where an SMPL template fits reliably, use SMPL-anchored Gaussians with linear blend skinning (LBS); far-range or template-less actors (cyclists, strollers), where the fit is too noisy to trust, are routed to a shared self-supervised deformation network. This routing is what lets OmniRe avoid conditioning far-range humans on an unreliable skeleton — a property central to the distance-dependence analysis in Section 5.1.3. Both mechanisms reintroduce the per-Gaussian computation Street Gaussians had avoided: a per-frame skinning blend in the SMPL case, and a per-Gaussian deformation-MLP pass in the deformable case.

OmniRe builds on SMPL, the skinned parametric model whose 24-joint skeleton also defines the rigid segments used here [Lop+15], and on linear blend skinning, which traces to classical pose-space deformation [LCF00]. SMPL-anchored Gaussians have become the standard testbed for animatable human avatars, and every such method couples Gaussians to the body model through learned skinning or deformation — SMPL-driven LBS with skinning refinement [HHL24], pose-conditioned non-rigid de-

formation for clothing [Qia+24], joint scene-and-human disentangling from monocular video [Koc+24], CNN-regressed pose-dependent Gaussian maps [Li+24b], and articulated templates with learnable forward skinning [Lei+24] — all descendants of animatable-NeRF work such as Neural Body [Pen+21]. All rely on continuous skinning or a deformation network; to our knowledge, no prior Gaussian-splatting method tests *purely rigid per-segment* articulation as a deliberate efficiency point, which is the gap this thesis targets.

This thesis proposes an alternative that stays strictly within Street Gaussians’ design philosophy. The observation is structural: although a walking person is non-rigid as a whole, it is composed of segments — torso, upper arms, forearms, head, thighs, shins — that are each approximately rigid, connected by joints. Articulated motion can therefore be modeled by applying Street Gaussians’ rigid-object pipeline *hierarchically*, with one rigid object per body segment rather than one per person. Each segment lives in its own local coordinate frame, and per-frame pose estimates from upstream body-pose estimation provide the rigid transformations that bring it into world space at render time. The rendering pipeline, rasterizer, loss functions, and adaptive density control are unchanged — pedestrians simply contribute multiple rigid objects instead of one.

The efficiency framing follows directly: each Gaussian belongs to exactly one segment and is transformed by exactly one rigid pose, identical to the vehicle case — no per-Gaussian skinning weights, no render-time blend, no neural network, and no storage for skinning machinery.

5.1.1 Method

Body-segment representation. Each pedestrian h is decomposed into S rigid body segments $\{s_1, \dots, s_S\}$. The decomposition follows the SMPL kinematic skeleton, with one segment per major bone in the skeletal tree. The thesis uses $S = 10$ segments (Figure 6): pelvis (containing the spine and torso Gaussians), head (including neck), left and right upper arm, left and right forearm (including hand), left and right upper leg, and left and right lower leg (including foot). Hands and feet are absorbed into their adjacent forearm and lower-leg segments respectively, due to limited pixel resolution at typical pedestrian distances.

Anchor joint selection. Each segment has one *anchor joint* from the SMPL 24-joint kintree that defines its local coordinate frame: pelvis, neck, left/right shoulder, left/right elbow, left/right hip, and left/right knee respectively. The anchor joints are placed at the elbow and knee rather than the wrist and ankle, because elbow and

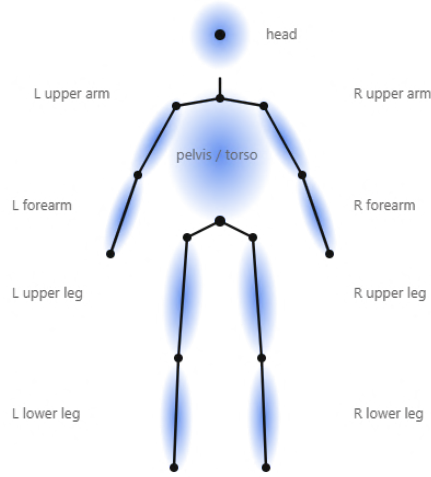


Figure 6 Part-based decomposition of a pedestrian into $S = 10$ rigid body segments. The skeleton marks each segment’s anchor joint; the blue point clouds indicate the disjoint sets of Gaussians assigned to each segment.

knee articulations are the ones that visibly fail in the rigid baseline B0 (gait cycle, arm swing) and are resolvable at typical Waymo pedestrian sizes ($\sim 80\text{--}150$ px tall at 10–30 m range). Wrist and ankle articulation is below the effective resolution limit of single-camera Waymo footage and would add segments with too few informative pixels to supervise.

Each segment s is represented as an independent set of 3D Gaussians $\bar{\mathcal{G}}_s = \{g_i^s\}$ in a local coordinate frame attached to the segment’s proximal joint. Each Gaussian g_i^s has the standard 3DGS attributes (position μ_i^s , rotation quaternion $\mathbf{q}_i^s$, scale $\mathbf{s}_i^s$, opacity α_i^s , and view-dependent color via spherical harmonics of degree $\ell_{\max} = 1$). The local frames are oriented along the segment’s principal axis, which makes the local geometry of the segment approximately constant across all poses — the local frame moves with the segment.

The SH degree of 1 is inherited from the original pipeline (Section 2.3), but is particularly important here: the decomposition introduces up to S times more Gaussians per pedestrian, so the fourfold appearance-storage cost of degree 3 would make memory the binding constraint at any realistic pedestrian count.

The total set of Gaussians representing pedestrian h is the union $\bar{\mathcal{G}}_h = \bigcup_s \bar{\mathcal{G}}_s$. No Gaussian belongs to more than one segment; the assignment is hard rather than weighted. This is the key simplification compared to OmniRe’s vertex-level skinning, where each Gaussian is anchored to a vertex on the SMPL mesh and is influenced by a weighted combination of joint transformations.

Per-frame transformation. At frame t , the SMPL pose estimate $\boldsymbol{\theta}(t)$ provides 24 axis-angle joint rotations describing the relative orientation of each segment with respect to its parent in the kinematic tree. Combined with the global pose $T_h(t) = (\mathbf{R}_h(t), \mathbf{t}_h(t))$ of the pelvis, forward kinematics on the SMPL skeleton yields the world-space pose $T_s(t) \in SE(3)$ of each segment s .

Given $T_s(t)$, each Gaussian in segment s is transformed into world coordinates by a single rigid-body operation, identical in form to Street Gaussians’ vehicle transformation:

$$\boldsymbol{\mu}_w^s = \mathbf{R}_s(t) \boldsymbol{\mu}_i^s + \mathbf{t}_s(t), \quad \mathbf{q}_w^s = \mathbf{R}_s(t) \otimes \mathbf{q}_i^s, \quad (5.1)$$

where $\otimes$ denotes quaternion composition. The world-space covariance follows from $\mathbf{q}_w^s$ and the unchanged scale $\mathbf{s}_i^s$ via the standard factorization — a single rigid transform per Gaussian, exactly as in the vehicle case.

Pose refinement. SMPL pose estimates from monocular RGB are noisy, particularly under occlusion or at large distances. Following Street Gaussians’ approach for vehicle poses, the thesis treats the segment poses as partially learnable. The world-space transformation in Equation 5.1 is replaced by

$$T'_s(t) = T_s(t) \cdot \Delta T_s, \quad (5.2)$$

where $\Delta T_s \in SE(3)$ is a learnable residual, parameterized as a rotation quaternion $\Delta \mathbf{q}_s$ (initialized to identity) and a translation vector $\Delta \mathbf{t}_s$ (initialized to zero). The rotation is composed with the data rotation: $\mathbf{R}_{\text{corrected}}[\text{anchor}_s] = \Delta \mathbf{R}_s \otimes \mathbf{R}_{\text{data}}[\text{anchor}_s]$.

Per-instance, not per-frame. Critically, the residuals are shared across all frames for a given pedestrian instance (*per-instance, per-segment*). Per-frame residuals would see one observation per parameter, inviting overfitting to photometric noise; per-instance residuals see every frame in which the pedestrian appears (up to the full 200-frame scene), so they absorb the *systematic* per-person bias in the 4D-Humans estimate (e.g. a consistently mis-estimated shoulder angle) rather than per-frame noise — analogous to Street Gaussians’ vehicle pose refinement. The total residual state is modest ($\approx 2,310$ parameters across 33 pedestrians), negligible relative to the Gaussian parameters. An L_2 penalty toward identity (weights 0.01; Appendix A.5) keeps corrections small unless the photometric loss strongly pulls them away. Gradients flow from the photometric loss through the rasterizer and rigid transformation into the residuals — the same direct path Street Gaussians exploits for vehicles. The residuals correct only *small* systematic errors; they cannot recover from a complete estimator failure (e.g. a missed detection), in which case the segment is omitted from that frame.

Initialization and density control. Each segment is initialized from the canonical SMPL template surface for the pedestrian’s shape parameters, with each template vertex assigned to a single segment by the argmax of its SMPL skinning weights and re-centered into that segment’s anchor-joint frame (details in Appendix A.5). The assignment is hard rather than weighted: unlike OmniRe, where each Gaussian is anchored to a mesh vertex and influenced by a blend of joint transforms, every Gaussian here belongs to exactly one segment. Adaptive density control then operates per segment rather than per pedestrian: densification and pruning use each segment’s local-frame view-space gradients, and Street Gaussians’ bounding-box pruning is applied in the segment’s local extent. This is what makes the design work — a Gaussian whose limb has swung outside the pedestrian’s overall bounding box is not pruned, because its segment’s local box has moved with it.

5.1.2 Experimental Setup

Scene and data. Experiments are conducted on several scenes from the Waymo Open Dataset training split, each 200 frames. The final evaluation is run on the DriveStudio repository, extended with the calibration fix (Appendix A.1) and the part-based pedestrian representation developed in this thesis. All scenes contain pedestrian instances with valid SMPL pose data obtained from the DriveStudio data bundle, which provides per-frame, per-instance SMPL pose quaternions and shape parameters estimated by 4D-Humans [Goe+23]. The estimates are monocular and per-frame, and are known to exhibit systematic per-person bias and per-frame noise, particularly for small or occluded pedestrians.

All scenes are trained with a single front-center camera at native resolution (**1-cam**). The 12 GB VRAM budget of the training hardware (Appendix A.3) does not accommodate a multi-camera configuration at native resolution; fitting three cameras within the same budget would require downscaling each, and the resulting per-camera resolution change would confound any comparison against the native-resolution single-camera runs. A multi-camera configuration is therefore not used, and all cross-method comparisons are made at identical 1-cam resolution.

Evaluation protocol. Every tenth frame is held out for evaluation (`test_image_stride=10`); all reported metrics are on these held-out frames. PSNR is computed on three regions: full image, pedestrian-only (projected 3D bounding boxes; see Appendix A.2 and Figure 11), and vehicle-only. The vehicle-only metric serves as a within-run sanity check, since all methods use identical `RigidNodes` for vehicles. Training details and hardware are described in Appendix A.3.

Methods compared. **(B0) Rigid baseline.** Each pedestrian routed to a single `RigidNodes` instance — the pipeline’s default when pedestrians are included in the dynamic object set.

(M) Part-based rigid decomposition (this thesis). Pedestrians decomposed into $S = 10$ rigid segments with per-segment learnable pose residuals, as described in Section 5.1.

(M-noref) Part-based without pose refinement. Identical to M but with pose residuals frozen at identity, isolating the contribution of the structural decomposition from that of the learned correction.

(OmniRe) External upper bound. OmniRe [Che+25] run with its full pipeline (the distance-based skinning/deformation routing described at the start of Section 5.1). Unlike B0, M, and M-noref, which share the same rigid codebase and differ only in the pedestrian representation, OmniRe is a separate method with its own implementation; it is included as a reference upper bound rather than as a controlled ablation, and its numbers are reported but not bolded in Table 4. It is evaluated on all four scenes.

Background and vehicle handling are identical across the B0, M, and M-noref methods.

5.1.3 Results

Method M decomposes pedestrians into a small number of rigid body segments posed via forward kinematics on a monocular-estimated SMPL skeleton. Across four Waymo scenes the result is consistent and distance-dependent: the part-based method helps where pedestrians are close enough that their articulation is resolvable, neither helps nor hurts much where they are not, and never diverges catastrophically. On the close-range scene 23 it improves Human-only PSNR by **+3.69 dB** over the single-rigid-body baseline B0 (1-cam), and on scene 552 by a further **+0.51 dB**; on the far-range-dominated scenes 114 and 327 the aggregate human metric instead favours B0 by 0.8 and 0.3 dB, because monocular SMPL estimation becomes unreliable at distance and M, unlike B0, commits to that noisy skeleton. The qualitative figures confirm the mechanism directly: M renders sharp, articulated limbs on close- and mid-range pedestrians where B0 smears them, and smears far-range pedestrians where B0 — being effectively rigid at sub-pixel limb motion — stays clean.

Three further findings cut across all scenes and are developed in the subsections below: the gain is almost entirely structural (the learned pose refinement contributes

only +0.01 to +0.38 dB, at or below run-to-run noise, so the decomposition, not the residual, is load-bearing); the representation is essentially free (it trains in the same wall-clock time as the rigid baseline, whereas OmniRe’s skinning-and-deformation pipeline trains 12–28% longer); and OmniRe remains the human-quality upper bound (~ 1 dB over M where both run cleanly), quantifying what the lightweight design forgoes. The picture is therefore a clear trade-off, not an unqualified win: M recovers a meaningful fraction of the close-range articulation B0 loses at no extra training cost, but where its pose prior fails it renders *below* the baseline and visibly smears the affected pedestrians — the aggregate margin stays small (under 0.8 dB) only because the human metric is diluted by shared static background, and M has no mechanism to decline a prior it cannot trust. The confidence-gated fallback of Section 5.2 that would bound this failure is left as future work, so the present method exhibits the trade-off as measured, not a recovered worst case.

Quantitative results. Table 4 reports PSNR and training wall-clock time on held-out test frames across all four scenes and both camera configurations.

Scene 23: Close-Range Pedestrians

On scene 23, which contains predominantly close-range pedestrians at 5–10 m from the ego vehicle, M substantially outperforms B0: Human PSNR improves by **+3.69 dB** ($25.70 \rightarrow 29.39$) in the 1-cam configuration. This is the strongest of the close-range results and reflects the central failure mode of the rigid baseline — a single rigid transform per pedestrian averages over independent limb motions during a walking cycle, producing smearing and ghosting, whereas the per-segment decomposition lets each limb track its own rigid trajectory, including per-leg articulation entirely absent from B0. OmniRe remains the upper bound at Human PSNR 30.12, but M trails it by only 0.73 dB while requiring neither skinning nor a render-time deformation network.

Figure 7 makes the failure mode and its resolution directly visible. In the B0 render, the legs — moving non-rigidly during the gait cycle — blur into a fan across the entire pedestrian bounding box, because a single rigid transform carries the whole point cloud along the trajectory and the swinging limbs cannot separate. The upper body renders comparatively cleanly, since the torso is approximately rigid and only displaced by the global pose. M, by contrast, is clean throughout: the legs appear as separated, articulated limbs with bending knees, and the leg ghosting of B0 is absent.

The single-camera regime in which this gain is measured is also the data-impoverted one most relevant to budget-constrained deployments: with only a front-center view, a

Table 4 Quantitative comparison on Waymo scenes 23, 114, 327, and 552, one camera at full resolution (1-cam), 30k iterations. All methods share the same evaluation protocol and resolution, so cross-method comparisons are controlled. Background and vehicle handling are identical across the B0/M-noref/M rows. “Wall” is end-to-end training wall-clock time. Best Human PSNR per scene in **bold**; OmniRe is not bolded as it is an external reference.

Scene	Method	Full PSNR $\uparrow$	Human PSNR $\uparrow$	Vehicle PSNR $\uparrow$	Wall (h)
23	B0	32.06	25.70	29.48	3.52
23	M-noref	34.27	29.01	29.31	3.49
23	M	33.96	29.39	29.42	3.50
23	OmniRe	35.28	30.12	30.19	4.17
114	B0	32.12	22.66	26.78	3.46
114	M-noref	31.95	21.87	26.60	3.44
114	M	31.93	21.88	26.59	3.45
114	OmniRe	34.91	26.74	27.78	4.18
327	B0	34.42	27.16	27.00	3.75
327	M-noref	34.61	26.73	26.62	3.73
327	M	34.65	26.86	26.91	3.73
327	OmniRe	35.65	30.78	21.18 [†]	4.17
552	B0	31.71	24.70	25.70	3.65
552	M-noref	31.99	25.20	25.53	3.33
552	M	32.07	25.21	26.18	3.27
552	OmniRe	32.34	26.21	25.98	4.18

single rigid cloud has the fewest geometric constraints to compensate for its averaging of limb motion, so the part-based decomposition has the most to add.

Structural decomposition vs. pose refinement. The M-noref ablation isolates the per-instance pose residuals: on scene 23, M-noref reaches Human PSNR 29.01 against M’s 29.39, a gap of **+0.38 dB**. So of the total 3.69 dB improvement over B0, the part-rigid representation driven by the per-frame forward-kinematics pose stream accounts for **≈3.31 dB**, while the trainable residuals add at most **0.38 dB** on top — and, on the basis of the single-run caveat above, that residual contribution is at or below the level of run-to-run noise on every scene. The load-bearing component of the method is therefore the structural decomposition posed by the 4D-Humans skeleton, not the learned correction the residuals provide; the residual is a minor, near-inert add-on rather than a second source of the gain. (Note that B0 $\rightarrow$ M-noref changes two things at once — the segmentation into rigid parts *and* the per-frame



(a) B0 (single rigid body)



(b) M (part-based rigid decomposition)

Figure 7 Qualitative comparison of a walking pedestrian on scene 23. **(a)** In the rigid baseline B0, the non-rigidly moving legs are smeared across the full pedestrian bounding box, while the approximately rigid upper body (displaced only by the global pose transform) renders comparatively cleanly. **(b)** The part-based decomposition M renders articulated legs as separated limbs with visibly bending knees and no ghosting. Per-leg articulation is the most compelling qualitative artifact of this thesis.

articulation that drives them — so the 3.31 dB is properly attributed to the FK-posed representation as a whole, not to the segmentation geometry in isolation.)

As a consistency check, Vehicle PSNR — which should be unaffected by the pedestrian model, since vehicles use identical `RigidNodes` across B0, M, and M-noref — varies by at most 0.65 dB across these rows within any scene (excluding OmniRe’s anomalous scene-327 vehicle value, discussed below). This stability is consistent with the rigid vehicle pipeline being genuinely unchanged across methods; it does not, however, constitute a variance estimate for the human metric, which all numbers in this thesis report from a single training run per cell without seed repetition. Sub-0.5 dB differences — including all of the M-versus-M-noref gaps and the far-range margins below — should therefore be read as within plausible run-to-run noise rather than as precise effects, and only the large close-range gap (+3.69 dB on scene 23) is comfortably beyond any plausible single-run variance.

Distance Dependence Across Scenes

The remaining three scenes resolve into a single mechanism, summarised here; the full per-scene numbers, figures, and failure-mode diagnosis are given in Appendix A.4.

Far range (scenes 114 and 327). Where pedestrians are small, the ordering inverts relative to scene 23: M falls **0.78 dB** below B0 on scene 114 (21.88 vs. 22.66) and **0.30 dB** below on the far-range-dominated scene 327 (26.86 vs. 27.16), with M-noref essentially tied with M in both. The cause is the SMPL prior, not scene difficulty: M

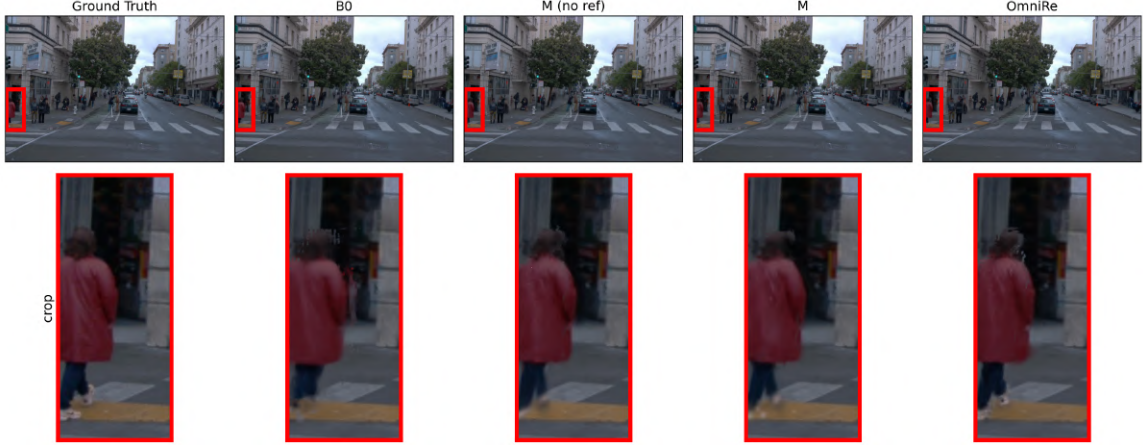


Figure 8 Mid-range pedestrian on scene 327, rendered by all four methods (columns: ground truth, B0, M-noref, M, OmniRe), with a zoomed crop below each frame. B0 shows mild residual leg ghosting; M-noref and M both resolve two distinct legs and are nearly indistinguishable from each other; OmniRe additionally recovers fine detail such as the shoes that the rigid-segment methods blur.

commits to a monocular skeleton whose joint-angle error grows sharply on 60–80 px subjects [Goe+23], so the ten segments scatter and smear (Figure 9), whereas B0 never reads the skeleton and stays clean at sub-pixel limb motion. OmniRe avoids this by routing far actors to its self-supervised deformation branch, holding a ~ 5 dB lead over M — direct evidence that declining to condition on the noisy skeleton is what matters. Scene 327 is decisive because it contains both ranges under identical capture conditions: M recovers the mid-range pedestrian (Figure 8) while the dominant far-range pedestrians smear (Figure 9), isolating distance as the operative variable. This unconditional commitment to the prior is the gap the confidence-gated fallback in Section 5.2 is designed to close.

Mixed range (scene 552). Scene 552 gives the clean four-method ordering $B0 < M\text{-noref} \approx M < \text{OmniRe}$, with **M +0.51 dB over B0** ($24.70 \rightarrow 25.21$). It complements scene 23 as a second scene where the decomposition helps in aggregate, closing about a third of the B0-to-OmniRe gap, with OmniRe’s full pipeline the upper bound at +1.00 dB over M. Pose refinement is inert here (M vs. M-noref, +0.01 dB), consistent with the cross-scene pattern that the decomposition, not the residual, is load-bearing.

A non-comparable number. One OmniRe entry must be flagged rather than read at face value: its scene-327 Vehicle PSNR (21.18, Table 4, marked $\dagger$) is a per-scene reconstruction fault in its separate pipeline — confirmed normal (25.98) on scene 552 — which the pooled Full PSNR inherits, so OmniRe’s scene-327 Full PSNR is not directly comparable to the rigid-codebase methods. Only its Human PSNR is comparable on that scene. This is why the OmniRe rows are reported for reference and not bolded.



Figure 9 Far-range pedestrians on scene 327, rendered by all four methods (columns: ground truth, B0, M-noref, M, OmniRe), with a zoomed crop below each frame. Relative to the mid-range pedestrian in the same scene (Figure 8), the ordering inverts: B0 is clean (limb motion is sub-pixel at this distance, so the rigid assumption effectively holds), whereas M and M-noref smear the pedestrians because they commit to a noisy far-range SMPL skeleton. OmniRe stays near ground truth, as it routes far-range actors to its self-supervised deformation branch rather than the SMPL path.

Training cost. The wall-clock column of Table 4 bears out the efficiency claim that motivates the design. On scene 327 (1-cam), B0, M-noref, and M all train in essentially the same time (3.75, 3.73, 3.73 hours), and on scene 552 again within minutes of each other (3.65, 3.33, 3.27 hours), with M and M-noref if anything marginally *faster* than B0, well within run-to-run variance. This is the central efficiency result: the ten-segment decomposition and per-segment residuals cost no measurable extra training time, because each Gaussian still undergoes exactly one rigid transform with no skinning blend or render-time network, and the $\sim 2,300$ residual parameters are negligible. OmniRe, by contrast, trains in 4.17 and 4.18 hours on these scenes — 12–28% longer — the expected price of its per-Gaussian skinning blend and deformation-network pass at every iteration. The gap is modest in absolute terms (under an hour) but captures the trade-off the thesis frames: OmniRe spends per-Gaussian computation to buy human-region quality, while the part-based method holds cost at the rigid-baseline level and recovers articulation through structure alone.

Limitations and qualitative artifacts. Two categories of visual artifacts were observed across all runs. First, the floater-mitigation overrides applied in this batch (depth-loss weight 0.05, tighter background cull thresholds) successfully suppressed the “Background Gaussian floater” artifact observed in initial experiments, where background Gaussians protruded visibly from vehicle sides.

A separate artifact remains: thin horizontal streaks (~ 5 px) appearing around pedestrian heads and vehicle edges, with brief gaps between consecutive streaks. This is consistent with anisotropic-Gaussian aliasing — individual highly elongated Gaussians viewed nearly edge-on project to thin streaks under the `gsplat` rasterizer (cur-

rent config: `antialiased: false`). The artifact appears on both `RigidNodes` vehicles and `PartRigidNodes` pedestrians, confirming it is a rasterizer-level phenomenon rather than one specific to the part-based design. It is precisely the failure that Mip-Splatting addresses by bounding each Gaussian’s frequency content to the sampling rate of the input views and replacing 3DGS’s screen-space dilation with a Mip filter [Yu+24]; enabling gsplat’s equivalent mip-style antialiasing would likely mitigate the streaks and is recommended as a first step in follow-up work.

Two artifacts are intrinsic to the part-based design and constitute the expected gap relative to a full LBS approach. Because foot Gaussians rigidly ride the knee joint’s transform, walking feet do not exhibit ankle pivot and appear slightly smeared during the stance-to-swing transition. Adjacent segments share no parameters at joint boundaries, producing faint discontinuities at the shoulder and hip — the kind of seam that vertex-level LBS smooths over.

5.2 Future Directions

The implementation in Section 5.1 addresses one specific limitation of Street Gaussians. This section discusses concrete directions for extending the part-based pedestrian representation and situates Street Gaussians within the broader research landscape.

5.2.1 Refining the Part-Based Pedestrian Representation

The part-based design in Section 5.1 occupies one design point in a larger trade-off space. Three concrete directions for further work are worth noting.

A *soft-boundary refinement* would relax the hard segment assignment at joints, where the one-segment-per-Gaussian rule produces visible discontinuities. Allowing Gaussians within a small radius of a joint to be transformed by a weighted blend of the two adjacent segment poses — localized LBS applied only at joint boundaries — recovers most of the smoothness benefit of full LBS at a fraction of the cost, since the skinning weights are non-zero for only a small fraction of Gaussians.

A *pose-conditioned appearance correction* would address the second remaining gap to OmniRe: pose-dependent appearance the rigid representation cannot capture. A small set of Fourier-style basis functions over the joint angles, producing per-Gaussian color residuals, could be added without a render-time MLP — closer in spirit to the original 4D SH formulation than to OmniRe’s residual MLP.

A *generalization to non-human articulated actors* (cyclists, scooters, strollers) would require a different decomposition rule, since these do not follow the SMPL skeleton; extracting part-level decompositions from off-the-shelf instance segmentation and applying the same rigid pipeline is one option, though its robustness to thin or unusual structures is an empirical question.

Confidence-gated fallback to a single rigid object. The scene-114 result (Section 5.1.3) exposes a structural weakness with a direct remedy: M commits to the SMPL skeleton unconditionally, yielding the +3.69 dB close-range gain when the estimate is reliable but rendering *below* B0 when it is noise, with no way to decline a prior that is wrong.

A natural fix is a *confidence-gated fallback*: route a pedestrian to the ten-segment `PartRigidNodes` representation only when the pose estimate is trustworthy, and otherwise to a single rigid `RigidNodes` object (the B0 behaviour). The gating variable is pose-estimate confidence — an occluded close-range pedestrian can be as unreliable as a distant one — proxied cheaply by projected bounding-box height (or range), optionally combined with the 4D-Humans per-joint uncertainty already in the pose bundle. This is a learning-free analogue of OmniRe’s `SMPLNodes/DeformableNodes` routing, selecting between part-rigid and single-rigid rather than between skinning and a deformation network; it adds no render-time cost and no new parameters. The worst case is bounded by construction: where the gate fires the method reduces to B0, so the 0.78 dB scene-114 deficit would by definition be recovered. The open question is the close-range side of the threshold — where to set the cutoff so the gate does not fire on pedestrians the decomposition would still improve — which scene 327 (Appendix A.4) is well suited to calibrate.

Adaptive temporal bandwidth for appearance. Appearance dominates per-Gaussian storage in the 4D SH formulation (60 floats versus 11 for geometry and opacity, Table 1), and the Fourier truncation order $k = 5$ is applied uniformly even though temporal variation is highly non-uniform across the body: the torso and head stay roughly constant over a crossing ($k_s = 1\text{--}2$ would suffice) while swinging limbs need the full $k_s = 5$. A heuristic that truncates any segment whose higher-order coefficients are near zero after a warm-up phase would exploit this; a hand-allocated assignment (Table 5) cuts per-pedestrian appearance storage roughly in half (a 48% reduction) with no expected quality loss on the truncated segments.

5.2.2 Feed-Forward Generalizable Gaussians

Street Gaussians requires 30 minutes of per-scene training with no knowledge transfer. The field has moved toward feed-forward Gaussian prediction to amortize this:

Table 5 Per-pedestrian appearance storage under three temporal bandwidth allocations. Each Gaussian stores $12 \times k_s$ appearance floats (4 SH basis functions $\times$ 3 color channels $\times k_s$ Fourier coefficients). Gaussian counts per segment are approximate, based on pixel coverage at typical pedestrian distances; the “adaptive” column assigns k_s per segment by expected temporal variation.

Segment	Gaussians	Uniform $k=5$	Adaptive k_s	k_s	Savings
Pelvis / torso	2,000	120,000	24,000	1	80%
Head	500	30,000	6,000	1	80%
L / R upper arm	2×300	36,000	14,400	2	60%
L / R forearm	2×200	24,000	14,400	3	40%
L / R upper leg	2×500	60,000	60,000	5	0%
L / R lower leg	2×400	48,000	48,000	5	0%
Total	5,300	318,000	166,800	$\bar{k} \approx 2.6$	48%

pixelSplat [Cha+24], MVSplat [Che+24], Splatter Image [SRV24], and GPS-Gaussian [Zhe+24] predict Gaussians in a single forward pass, but none decomposes into background plus tracked dynamic objects. DrivingRecon [Lu+25] is the first feed-forward 4D model for driving, predicting scene Gaussians including dynamic offsets from a surround-view sequence, generalizing to unseen Waymo scenes (+2.37 dB over pixelSplat) with transferable pre-trained representations. The open gap is that it runs at low resolution (256×512) on noisy auto-labeled masks; integrating Street Gaussians’ explicit object-track decomposition into a feed-forward architecture remains unsolved.

5.2.3 Annotation-Free Decomposition

Street Gaussians depends on 3D bounding-box annotations to separate dynamic objects from the background, limiting it to platforms with mature detection pipelines. Two recent methods remove this supervision. S³Gaussian [Hua+24] decomposes the scene with a multi-resolution HexPlane encoder, learning motion only where the photometric loss requires it, and on Waymo matches annotation-dependent methods on full-scene PSNR while exceeding them on dynamic-object PSNR (25.78 vs. 25.46 dB) using only RGB and LiDAR. DeSiRe-GS [Pen+25] instead derives a dynamicness map from a frozen DINOv2 to regularize a 3D velocity field, with geometric regularizers that improve surface reconstruction beyond Street Gaussians or OmniRe. Both evolve the scene-graph design directly: the decomposition principle is preserved, the annotation requirement is not.

6 Conclusion

Street Gaussians reconstructs dynamic driving scenes at state-of-the-art quality and real-time speed by treating each moving actor as a rigid object, an assumption that excludes articulated pedestrians; subsequent work restored them only by reintroducing the per-Gaussian skinning and deformation networks the pipeline was designed to avoid. This thesis asked whether pedestrians can instead be recovered *within* the rigid-object design, by decomposing each into ten approximately rigid body segments posed by forward kinematics on a SMPL skeleton — a pedestrian contributing ten rigid objects rather than one, with no skinning, no blending, and no render-time network.

On four Waymo scenes the answer is a qualified yes, and the qualification is the main result. Where pedestrians are close, the decomposition improves human-region PSNR by up to 3.69 dB over a single-rigid-body baseline at no measurable training cost, removing the limb smearing a single transform produces over the gait cycle; an ablation attributes the gain to the structural decomposition rather than the learned pose residual. The benefit is distance-dependent, however, and the adverse half deserves equal weight: where monocular SMPL estimates degrade at far range, committing to a noisy skeleton scatters the segments and drives the decomposition up to 0.78 dB *below* the baseline, which never reads the skeleton and is immune to its errors. The structured representation amplifies pose-estimation error rather than averaging it out, precisely because it cannot decline a prior that has become untrustworthy. These results speak to the four scenes evaluated and to the single-camera, budget-constrained regime they represent; they do not establish behaviour across datasets, denser camera rigs, or actor classes beyond pedestrians.

The clearest next step follows directly from that failure mode: a confidence-gated fallback that uses the ten-segment representation only when the pose estimate is trustworthy and a single rigid object otherwise. It adds no render-time cost and no new parameters, and its worst case is bounded by construction — where the gate fires the method reduces to the rigid baseline, recovering the far-range deficit by definition. More broadly, the field is moving past per-scene optimization and bounding-box supervision toward feed-forward and annotation-free reconstruction; carrying Street Gaussians’ explicit object-track decomposition into those settings, rather than discarding it, remains the open and worthwhile problem.

A Appendix

A.1 Waymo Camera Distortion

Waymo’s vehicle-mounted cameras are not pinhole sensors: each provides five OpenCV plumb-bob distortion coefficients alongside the standard focal-length and principal-point parameters, producing pixel displacements that grow toward the image corners.

In both the official and unofficial implementations, the COLMAP bootstrapping step writes its camera file using a pinhole model that drops the distortion fields, so the downstream undistortion path receives no coefficients and silently becomes a no-op. The fix is to switch the COLMAP camera model to `OPENCV` and forward the distortion coefficients that were already extracted upstream — a one-line preprocessing change with no modifications to the training code.

The empirical effect on reconstruction PSNR is modest but consistent across cameras, removing a systematic bias that would otherwise cause all subsequent experiments to underreport reconstruction quality. A secondary issue was also found in the side cameras, where the principal point c_y was offset by 197 pixels due to a double crop-correction; restoring the calibrated value from the Waymo intrinsics fixes this.

The broader takeaway is methodological: distortion handling is a silent preprocessing detail that can shift absolute PSNR numbers without producing any visible signal in training loss curves, and is best caught by reading the data flow rather than waiting for a visible failure mode.

A.2 Human PSNR Evaluation Region

The pedestrian-only metric Human PSNR reported throughout Section 5.1 follows the same construction as the PSNR* of the original Street Gaussians paper [Yan+24a], restricted to pedestrian boxes only: each pedestrian’s 3D tracked box is projected into the image plane, and the MSE is accumulated only over the pixels inside the resulting boxes. Because the region is the projected bounding box rather than a tight silhouette, some background pixels inside the box are still included in the metric; it nonetheless concentrates the score on the pedestrians instead of letting the static background that fills most of the frame dominate it.

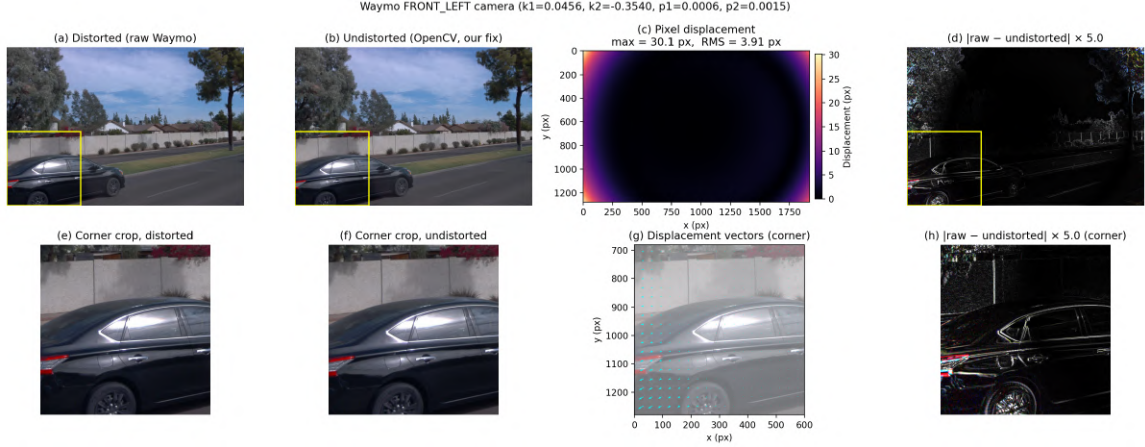


Figure 10 Effect of OpenCV distortion correction on the Waymo front-left camera. Top row: full-image comparison and per-pixel distortion magnitude. Bottom row: corner crop showing geometric displacement of a parked vehicle, with the difference image overlaid by displacement vectors. The distortion is centered around zero in the image middle and grows radially outward, consistent with the radial-distortion model.

A.3 Training Configuration and Hardware

All experiments run on a shared university JupyterHub cluster, on an NVIDIA A40 GPU constrained to a 12 GB VRAM allocation, with 64 GB system RAM and 4 CPU cores. Because only a fraction of the A40’s compute is allocated per session, absolute training times are slower than the card’s full throughput would suggest and should be read as relative comparisons between methods rather than as the hardware’s peak performance. The 12 GB VRAM cap is the binding constraint behind the single-camera (1-cam) protocol of Section 5.1: it does not accommodate a multi-camera configuration at native resolution, and downscaling to fit would change per-camera resolution and confound any comparison against the native-resolution runs. Memory management requires `data.preload_device=cpu` throughout. Each run trains for 30,000 iterations with the Adam optimizer and DriveStudio default learning-rate schedules. Three floater-mitigation overrides are applied identically to all runs: depth-loss weight 0.05, background `cull_alpha_thresh=0.01`, and background `cull_scale_thresh=0.3`. Individual runs take roughly 3.3–4.2 hours (Table 4), with the rigid-codebase methods at the low end and OmniRe at the high end. The final evaluation uses the DriveStudio repository, extended with the calibration fix described in Section A.1 and the part-based pedestrian representation developed in this thesis.

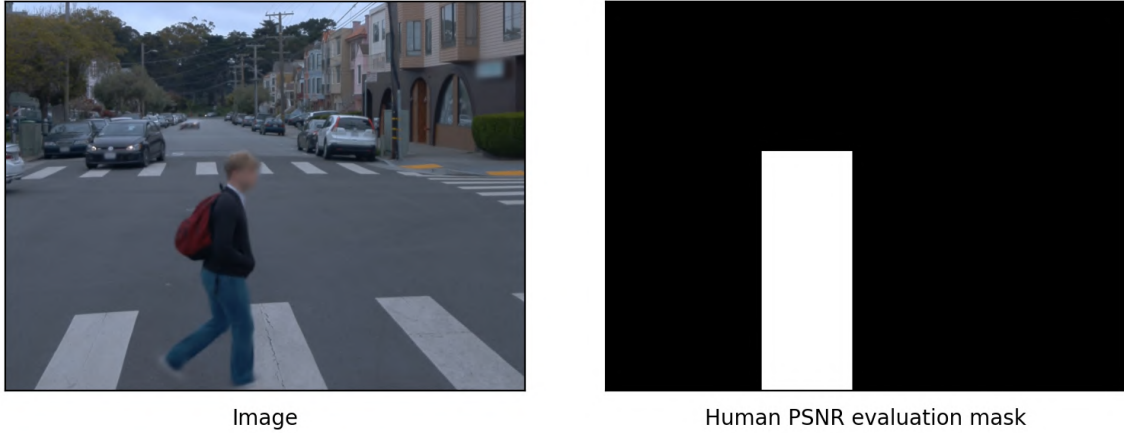


Figure 11 Illustration of Human PSNR. We project the 3D tracked pedestrian boxes to the 2D image plane and obtain the mask on the right. The MSE is computed only over the pixels within the mask to obtain the Human PSNR value.

A.4 Per-Scene Result Details

This section expands the distance-dependence summary of Section 5.1.3 with the full per-scene reasoning. All numbers are quoted from Table 4; the supporting figures (Figures 8 and 9) appear in the main text.

Scene 114: Far-Range Pedestrians

On scene 114, M does not improve over B0. Human PSNR is **0.78 dB lower** for M than B0 (21.88 vs. 22.66), and M-noref is essentially tied with M (21.87). Both methods score roughly 3–4 dB lower in absolute Human PSNR on scene 114 than on scene 23, confirming that scene 114 is intrinsically harder. OmniRe also pays some of this penalty — its Human PSNR falls to 26.74 here, 3.38 dB below its scene-23 figure of 30.12 — confirming that far range is hard for every method. But it degrades far more gracefully than M, holding a 4.86 dB lead over M’s 21.88, because its distance-based routing sends these far-range actors to the self-supervised deformation branch rather than the unreliable SMPL path. That contrast is strong evidence that M’s collapse is driven by the SMPL prior, not by scene difficulty alone: a method that declines to condition on the noisy far-range skeleton loses only a few dB where M loses many.

The diagnosis explains why the structured method underperforms the unstructured one precisely where structure ought to help least. Two effects compound. First, monocular pose estimation degrades on small subjects: 4D-Humans’ SMPL estimates carry substantially larger joint-angle error on pedestrians whose bounding boxes are only 60–80 px tall [Goe+23], and scene 114 pedestrians are predominantly at longer range than scene 23. M does not merely tolerate this noisy skeleton — it *commits* to

it, splitting each pedestrian into ten segments and rigidly placing every segment by its estimated per-frame joint pose. When those joint poses are wrong, the segments are scattered to wrong world positions every frame, and the per-segment clouds smear across the union of their erroneous placements. B0 is structurally immune to this failure because it never reads the skeleton in the first place. Second, the per-Gaussian photometric gradient signal collapses for small pedestrians: instances contributing only a few thousand pixels of loss across all frames divide that signal across $\sim 4,000$ Gaussians per pedestrian — and, in M, across ten segments — leaving densification, colour refinement, and the per-segment pose residuals with too little information to correct the bad initialisation. The structural prior thus amplifies pose-estimation error rather than averaging it out, which is why M falls *below* B0 here rather than merely failing to improve.

This contrast is exactly what OmniRe’s distance-based routing (Section 5.1) is designed to avoid. OmniRe does not force far-range pedestrians through its SMPL/LBS path; it routes them to a self-supervised deformation field learned directly from the images, which never conditions on the unreliable far-range skeleton. The part-based method M has no such fallback — it has only the SMPL-driven path — which is the architectural gap that the soft-fallback direction in Section 5.2 is meant to close.

Scene 327: Within-Scene Distance Dependence

Scenes 23 and 114 establish the distance dependence across two scenes, but they differ in more than pedestrian range. Scene 327 isolates the effect within a single scene: it contains pedestrians at both mid range and far range under identical capture conditions, lighting, and background. It is, however, dominated by far-range pedestrians — the close-range woman in Figure 8 is one instance among many distant ones — and this is decisive for reading the aggregate numbers in Table 4.

The aggregate Human PSNR on scene 327 (1-cam) does *not* favour the part-based method: B0 scores 27.16, M 26.86, and M-noref 26.73. B0 wins by 0.30 dB over M. This is the opposite of the scene-23 result and must be reported plainly: because the scene is far-range-dominated, the per-pixel human metric is driven by the many distant pedestrians where M loses (the scene-114 failure mode), and the single well-reconstructed mid-range pedestrian in Figure 8 is too small a fraction of the human pixels to move the average. Scene 327 therefore behaves quantitatively like scene 114, not scene 23 — it is further evidence for the distance dependence, not a counterexample to it. Two secondary observations survive the aggregate: M beats M-noref by +0.13 dB, consistent with the small positive contribution of pose refinement seen on scenes 23 and 114; and the gap between B0 and M is much smaller here (0.30 dB)

than the catastrophic far-range gap on scene 114 (0.78 dB), reflecting that 327 is a mix of distances rather than uniformly far.

With that caveat established, the figures show what the aggregate cannot: how each *individual* pedestrian is rendered at a given distance. Figure 8 shows the mid-range pedestrian. For this instance B0 still exhibits mild residual leg ghosting, while M-noref and M resolve two distinct legs; the visible difference between M-noref and M is small, consistent with the +0.13 dB aggregate gap. OmniRe is the sharpest, recovering low-level detail such as the shoes that the rigid-segment methods blur away — the residual quality the part-based design deliberately forgoes by avoiding vertex-level skinning. This per-instance advantage for M at mid range is real but local; it does not survive aggregation because the scene’s pixel budget is dominated by the far-range pedestrians in Figure 9.

Figure 9 shows far-range pedestrians from the *same* scene, and the per-instance ordering inverts. B0 is now very good: the pedestrians span so few pixels that their limb motion is sub-pixel and therefore effectively rigid in image space, so a single rigid cloud — which never consults a skeleton — reconstructs them cleanly. M and M-noref, by contrast, smear the pedestrians badly, dispersing them into an indistinct blur, exactly as the scene-114 diagnosis predicts: committing to a noisy far-range SMPL skeleton scatters the segments. Because these far-range instances dominate the scene, this is also what drives B0’s win in the aggregate human metric above. OmniRe is again the closest to ground truth on the humans (Human PSNR 30.78, the highest of any method on any scene in Table 4), because its distance-based routing sends these far-range actors to the self-supervised deformation branch rather than the SMPL path. Seeing both behaviours in one scene rules out scene-specific confounds and makes the distance dependence — not some other difference between scenes 23 and 114 — the operative variable.

One number in OmniRe’s scene-327 row must be flagged rather than read at face value: its Vehicle PSNR is 21.18 dB (Table 4, marked [†]), roughly 6 dB below every other method’s ~27 dB on the same scene. Vehicles are handled identically across the B0/M-noref/M rows, so this is not a property of the scene but of OmniRe’s separate pipeline — most likely a vehicle that failed to reconstruct or a difference in how its vehicle evaluation region is defined. Because the Full PSNR pools human, vehicle, and background pixels, OmniRe’s full-image 35.65 inherits this defect from the opposite direction (its strong human and background reconstruction offsets the broken vehicle), so its Full PSNR is *not* directly comparable to the others on scene 327. Only OmniRe’s Human PSNR on this scene should be compared across methods, and

even that comparison is against an external pipeline rather than a controlled ablation. This is exactly why the OmniRe rows are reported for reference and not bolded.

This is not a failure to acknowledge defensively — it is a finding to characterize. A method that conditions on monocular SMPL pose estimation will follow that estimate wherever it leads, including into error when the estimate is unreliable; without a confidence-gated fallback it cannot degrade gracefully to the unstructured baseline. The fact that M nonetheless stays within 0.3 dB of B0 on scene 327 (and within 1 dB on scene 114) shows the aggregate damage is bounded even when the scene is far-range-dominated, but the qualitative smearing makes clear that a fallback to a single rigid object (or a deformation branch) under low pose confidence is the right remedy.

Scene 552: A Clean Four-Method Ordering

Scene 552 is the most well-behaved of the four scenes, and the only one on which all four methods — B0, M-noref, M, and OmniRe — produce the textbook ordering with no anomalies. On Human PSNR, B0 scores 24.70, M-noref 25.20, M 25.21, and OmniRe 26.21. Two things are worth drawing out. First, **M improves on B0 by +0.51 dB** ($24.70 \rightarrow 25.21$): unlike the far-range-dominated scene 327, this scene contains enough mid-range pedestrian signal for the part-based decomposition to help in aggregate, so it complements scene 23 as a second scene where structure pays off. Second, the full ordering $B0 < M\text{-noref} \approx M < \text{OmniRe}$ holds exactly as the design narrative predicts: the rigid baseline is worst, the part-based method closes about a third of the B0-to-OmniRe gap (+0.51 of 1.51 dB), and OmniRe’s full skinning-and-deformation pipeline remains the upper bound at +1.00 dB over M.

Pose refinement is essentially inert on this scene: M beats M-noref by only +0.01 dB (25.21 vs. 25.20). This is consistent with the pattern across all scenes — refinement contributes at most a few tenths of a dB and sometimes nothing — and reinforces that the structural decomposition, not the learned residual, is the load-bearing component of the method.

Scene 552 also resolves the scene-327 vehicle anomaly. Here OmniRe’s Vehicle PSNR is 25.98, squarely in the normal range and within 0.5 dB of the rigid-codebase methods (B0 25.70, M-noref 25.53, M 26.18). Since OmniRe’s vehicle reconstruction is well-behaved on this scene, the broken 21.18 dB on scene 327 is confirmed as a per-scene reconstruction fault rather than a systematic property of the OmniRe pipeline, and the full-image and vehicle comparisons on scene 552 can be read at face value.

A.5 Part-Based Pedestrian: Implementation Details

This section collects the implementation specifics of the part-based pedestrian representation (Section 5.1) that are not needed to follow the method.

Pose-residual regularization. The per-instance, per-segment residuals are constrained by a quadratic penalty toward identity, keeping corrections small unless the photometric loss strongly pulls them away:

$$\mathcal{L}_{\text{reg}} = w_{\text{rot}} \overline{(1 - (\Delta q_{h,s}^w)^2)} + w_{\text{trans}} \overline{\|\Delta \mathbf{t}_{h,s}\|^2}, \quad (\text{A.1})$$

with $w_{\text{rot}} = w_{\text{trans}} = 0.01$, where $\Delta q_{h,s}^w$ is the scalar component of the residual quaternion and $\overline{(\cdot)}$ denotes the mean over all instances h and segments s . The rotation term $1 - (\Delta q^w)^2$ is a monotonic penalty that is zero at identity ($\Delta q^w = 1$) and increases with the residual rotation angle; for the small residuals that the weight enforces it is proportional to θ^2 , i.e. to $\|\mathbf{R} - \mathbf{I}\|^2$. The residual tensors have shape (N_instances, 10, 4) for rotations and (N_instances, 10, 3) for translations.

Initialization. Each pedestrian’s Gaussians are initialized from the canonical SMPL template surface, reusing the same template-vertex initialization as the parent single-rigid SMPL node. The 6,890 vertices of the SMPL mesh, instantiated in the canonical (rest) pose for the pedestrian’s shape parameters β , provide the initial Gaussian positions. Each vertex is assigned to exactly one of the ten segments by the argmax of its SMPL skinning weights over the segment’s constituent joints, yielding the disjoint per-segment partition $\bar{\mathcal{G}}_s$. The vertices of each segment are then re-centered into that segment’s anchor-joint frame, so that each segment’s initial Gaussians lie in its own local coordinate frame as required by the per-frame transformation. This template-based, skinning-partitioned initialization is fully reproducible and shape-adaptive through β , and gives per-segment adaptive density control a geometrically faithful starting point without ingesting per-frame LiDAR.

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